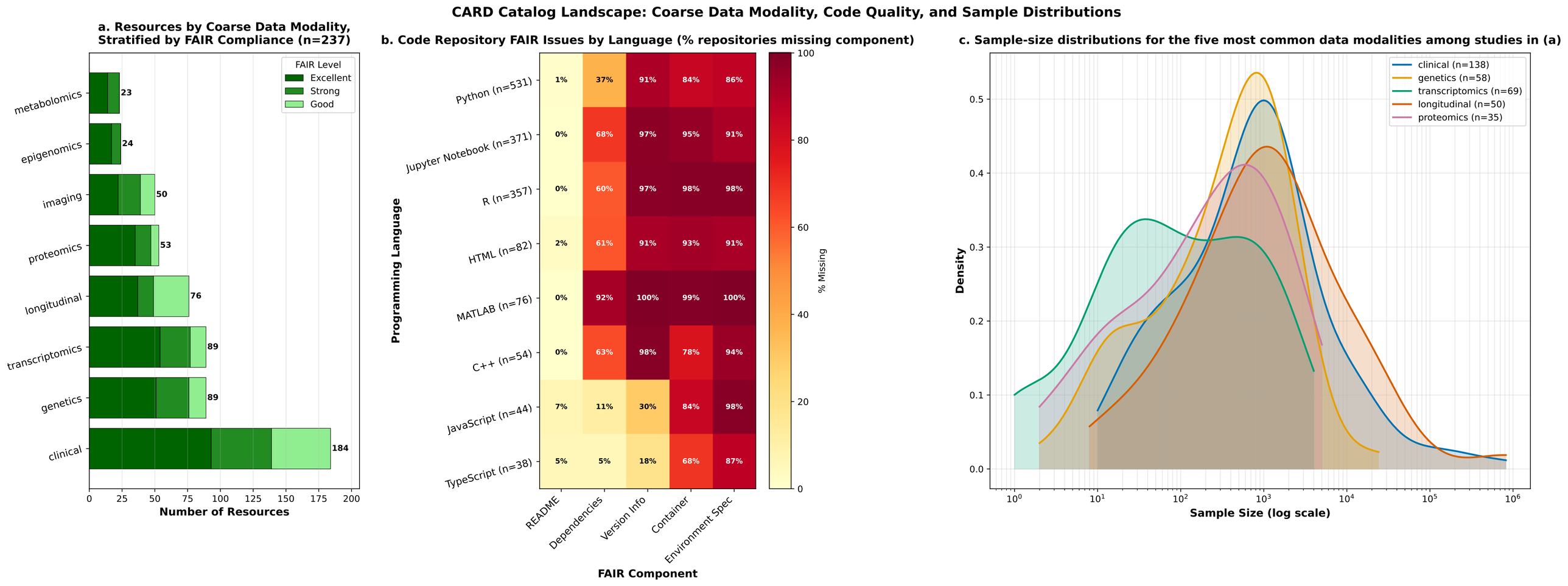

CARD Catalog Landscape: Coarse Data Modality, Code Quality, and Sample Distributions
a. Resources by Coarse Data Modality, Stratified by FAIR Compliance (n=237)
FAIR Level
Excellent
Strong
Good
metabolomics
epigenomics
imaging
proteomics
longitudinal
transcriptomics
genetics
clinical
23
24
50
53
76
89
89
184
0
25
50
75
100
125
150
175
200
Number of Resources
b. Code Repository FAIR Issues by Language (% repositories missing component)
Programming Language
Python (n=531)
Jupyter Notebook (n=371)
R (n=357)
HTML (n=82)
MATLAB (n=76)
C++ (n=54)
JavaScript (n=44)
TypeScript (n=38)
1% 37% 91% 84% 86%
0% 68% 97% 95% 91%
0% 60% 97% 98% 98%
2% 61% 91% 93% 91%
0% 92% 100% 99% 100%
0% 63% 98% 78% 94%
7% 11% 30% 84% 98%
5% 5% 18% 68% 87%
README
Dependencies
Version Info
Container
Environment Spec
FAIR Component
% Missing
0
20
40
60
80
100
c. Sample-size distributions for the five most common data modalities among studies in (a)
clinical (n=138)
genetics (n=58)
transcriptomics (n=69)
longitudinal (n=50)
proteomics (n=35)
Density
0.0
0.1
0.2
0.3
0.4
0.5
10^0
10^1
10^2
10^3
10^4
10^5
10^6
Sample Size (log scale)